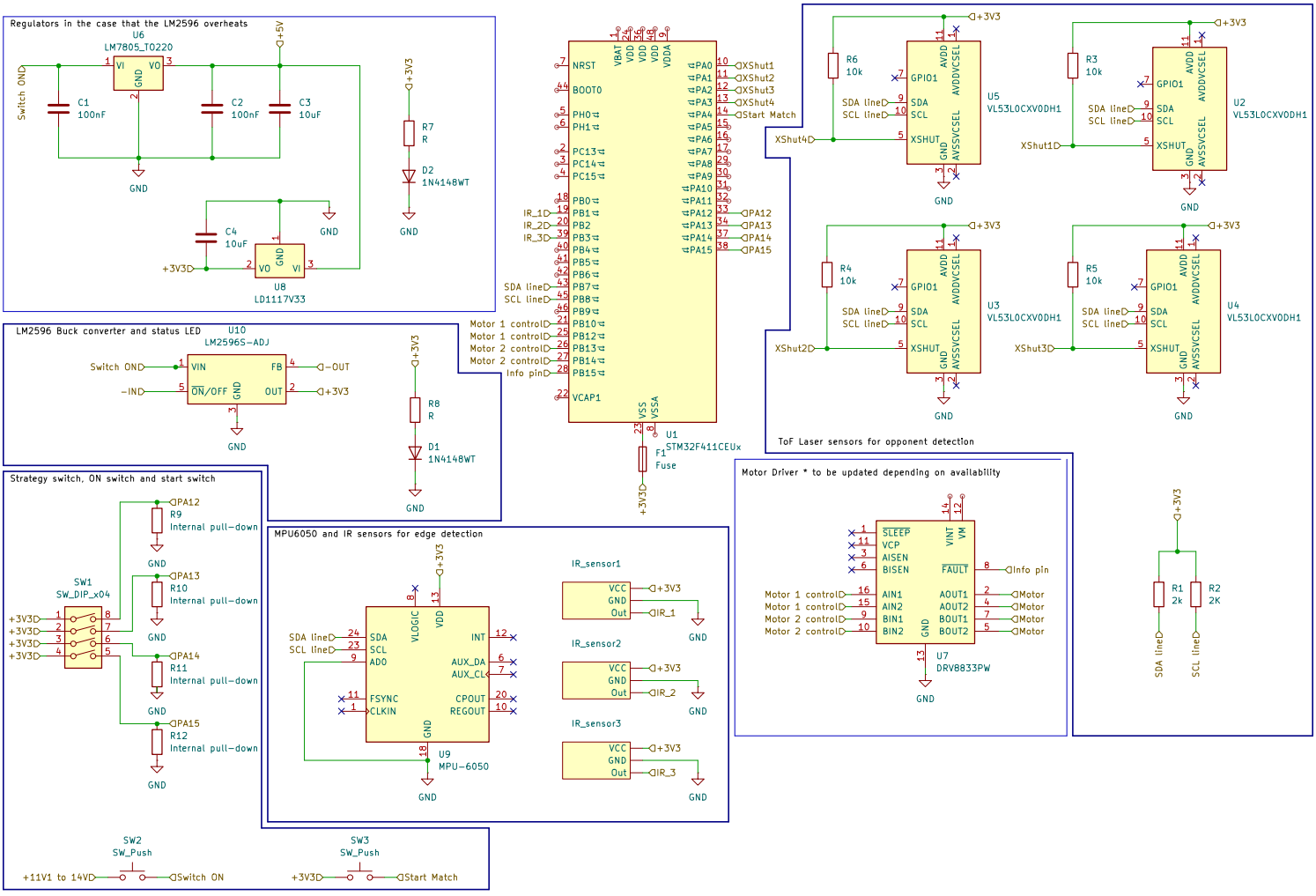

Regulators in the case that the LM2596 overheats
U6
LM7805_TO220
Switch OND
+5V
C1
100nF
C2
100nF
C3
10uF
GND
C4
10uF
+3V3
U8
LD1117V33
R7
R
D2
1N4148WT
LM2596 Buck converter and status LED
U10
LM2596S-ADJ
Switch OND
-IND
-OUT
VIN
FB
ON/OFF
OUT
R8
R
D1
1N4148WT
Strategy switch, ON switch and start switch
SW1
SW_DIP_x04
PA12
PA13
PA14
PA15
R9
R10
R11
R12
Internal pull-down
SW2
SW_Push
+11V1 to 14V
Switch ON
SW3
SW_Push
+3V3
Start Match
MPU6050 and IR sensors for edge detection
U9
MPU-6050
SDA line
SCL line
VLOGIC
VDD
INT
AUX_DA
AUX_CL
FSYNC
CLKIN
CPOUT
REGOUT
IR_sensor1
IR_sensor2
IR_sensor3
VCC
GND
Out
IR_1
IR_2
IR_3
U1
STM32F411CEUx
F1
Fuse
VBAT
VDD
VDDA
NRST
BOOT0
PH0
PH1
PC13
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
VCAP1
VSS
VSSA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
XShut1
XShut2
XShut3
XShut4
Start Match
IR_1
IR_2
IR_3
Motor 1 control
Motor 1 control
Motor 2 control
Motor 2 control
Info pin
ToF Laser sensors for opponent detection
U5
VL53L0CXV0DH1
U2
VL53L0CXV0DH1
U3
VL53L0CXV0DH1
U4
VL53L0CXV0DH1
R6
R3
R4
R5
10k
GPIO1
SDA
SCL
XSHUT
AVDD
AVDDVCSEL
AVSSVCSEL
Motor Driver * to be updated depending on availability
U7
DRV8833PW
SLEEP
VCP
AISEN
BISEN
VINT
VM
FAULT
AIN1
AIN2
BIN1
BIN2
AOUT1
AOUT2
BOUT1
BOUT2
Motor
R1
2k
R2
2K